Why are we here?
and other important questions

Bhaskaran Raman

Department of CSE, IIT Bombay

https://www.cse.iitb.ac.in/~br

Why are we here?

- What is LITERACY?
- What is KNOWLEDGE?
- What is EDUCATION?
- What is ENGINEERING/SCIENCE EDUCATION?

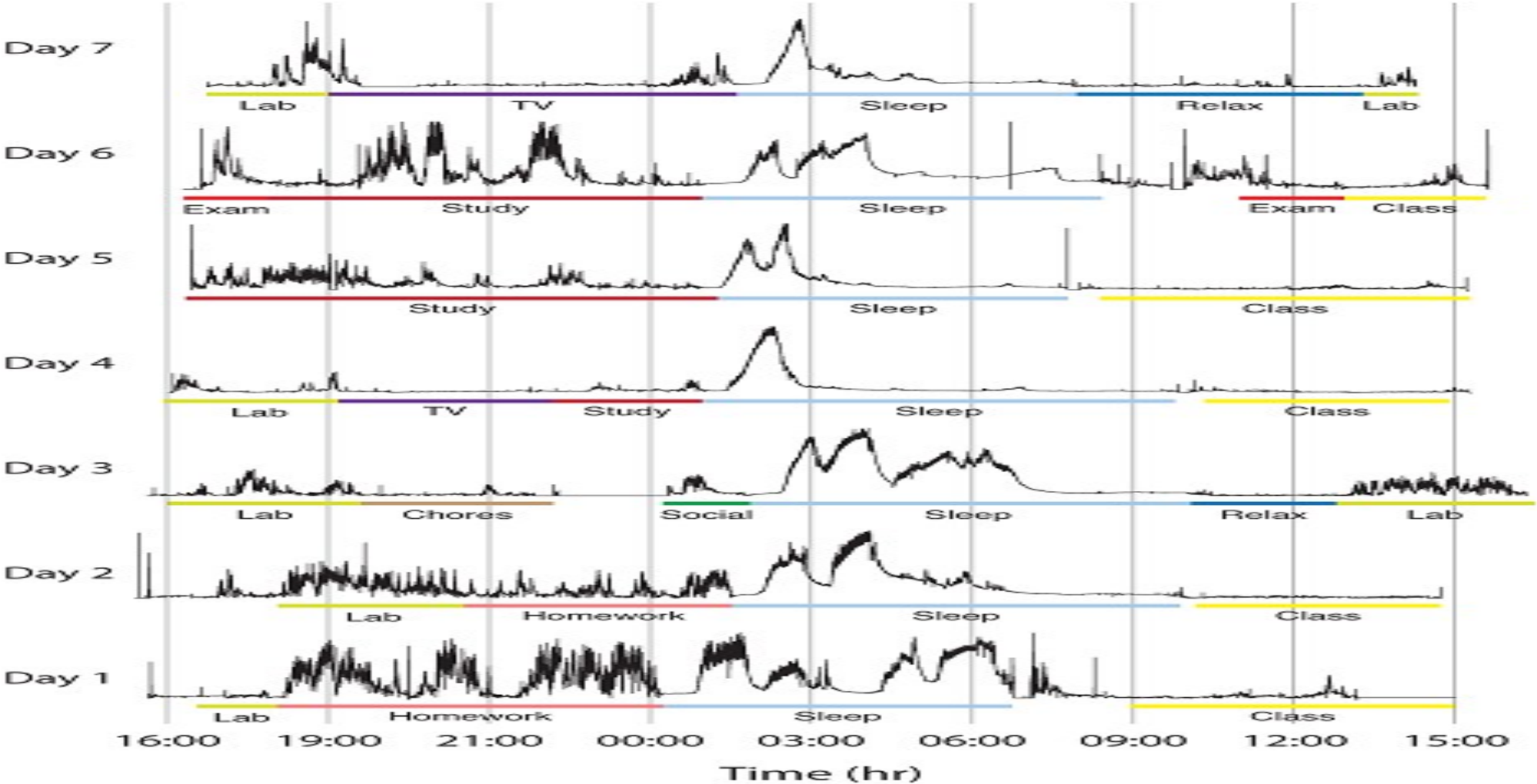
How do we learn?

A Word Game

- Ocean/breeze
- Leaf/tree
- Sweet/sour
- Movie/actor
- Fuel/engine
- Bread/b_tter
- Music/l_rics
- Sh_e/socks
- Phone/_all
- Pen_il/paper
- River/b_at
- Wine/g_ass
- Television/rad_o
- L_nch/dinner
- Tea_her/student
- School/college
- Fruit/vegetable
- Computer/memory
- Chair/table
- Bird/nest

From: “The Talent Code: Greatness Isn’t Born, Its Grown”, by Daniel Coyle

Electrodermal Activity



Reference: Poh, M.Z., Swenson, N.C., Picard, R.W., **May 2010,** doi: 10.1109/TBME.2009.2038487

Why am _I_ here?

Course Model

- Flipped Classroom: http://en.wikipedia.org/wiki/Flip_teaching

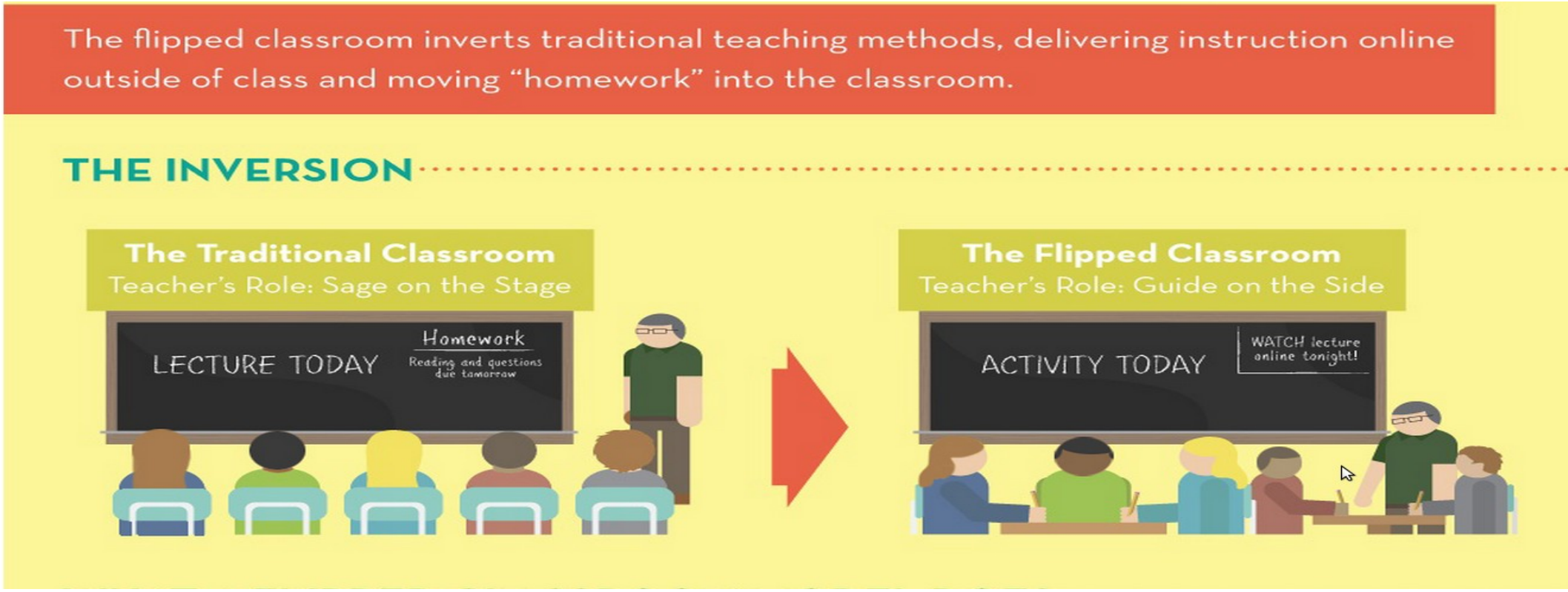


Fig. From http://www.knewton.com/flipped-classroom/

Opium for the masses

“Brain Drain”

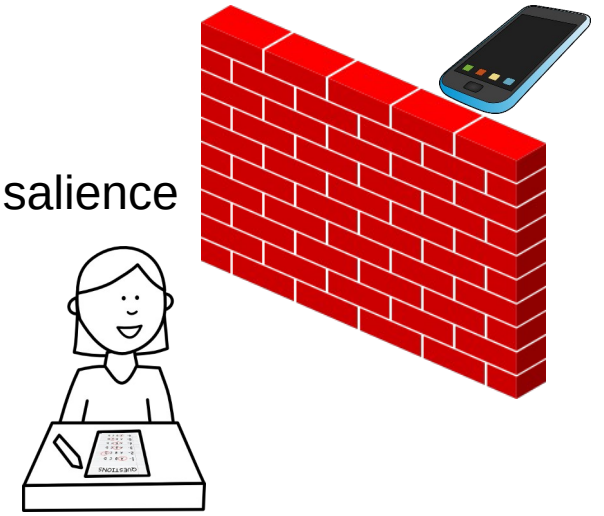
High salience



Medium salience



Low salience



“Our data indicate that the mere presence of one’s smartphone adversely affects two domain-general measures of cognitive capacity—available working memory capacity (WMC) and functional fluid intelligence (Gf)—even when participants are not using their phones and do not report thinking about them”

Ward, Adrian F., et al. "Brain drain: The mere presence of one’s own smartphone reduces available cognitive capacity." Journal of the Association for Consumer Research 2.2 (2017): 140-154.

Major Depression Among Teens

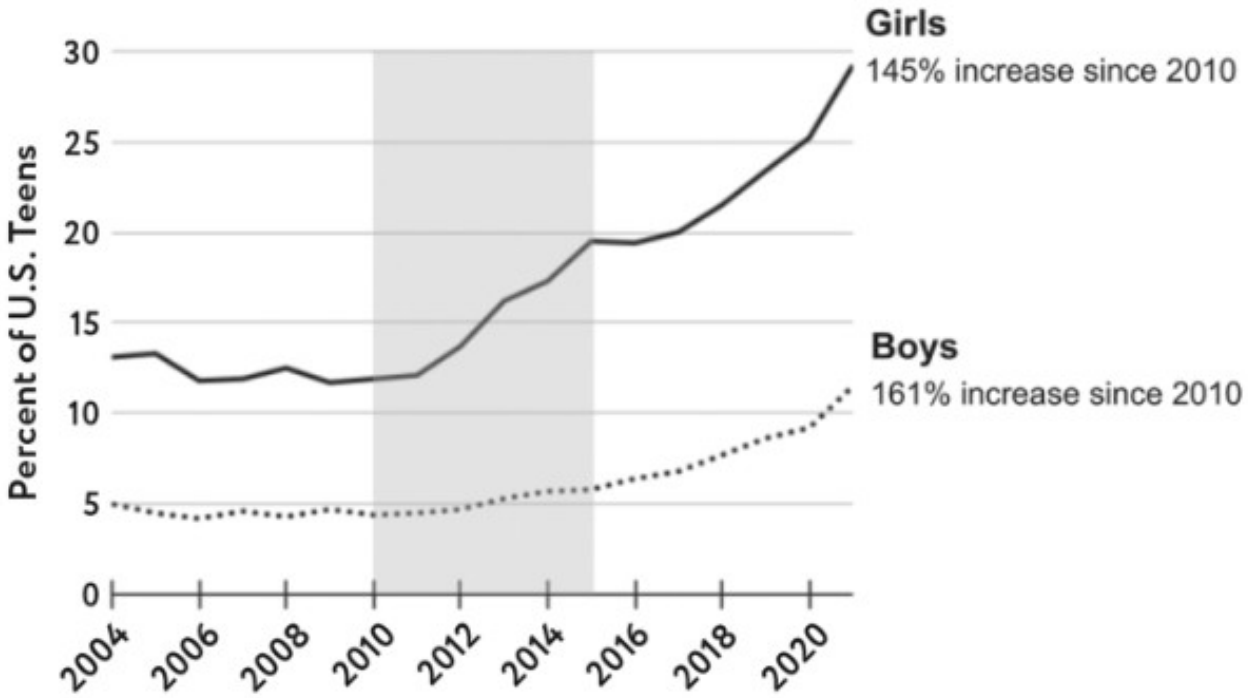
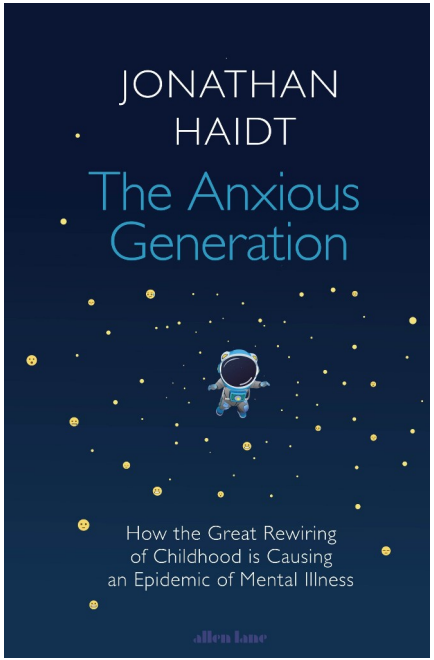


Figure 1.1. Percent of U.S. teens (ages 12–17) who had at least one major depressive episode in the past year, by self-report based on a symptom checklist. This was figure 7.1 in *The Coddling of the American Mind*, now updated with data beyond 2016. (Source: U.S. National Survey on Drug Use and Health.)[3]



"The Anxious Generation", by Jonathan Haidt

Mental Illness: College Students

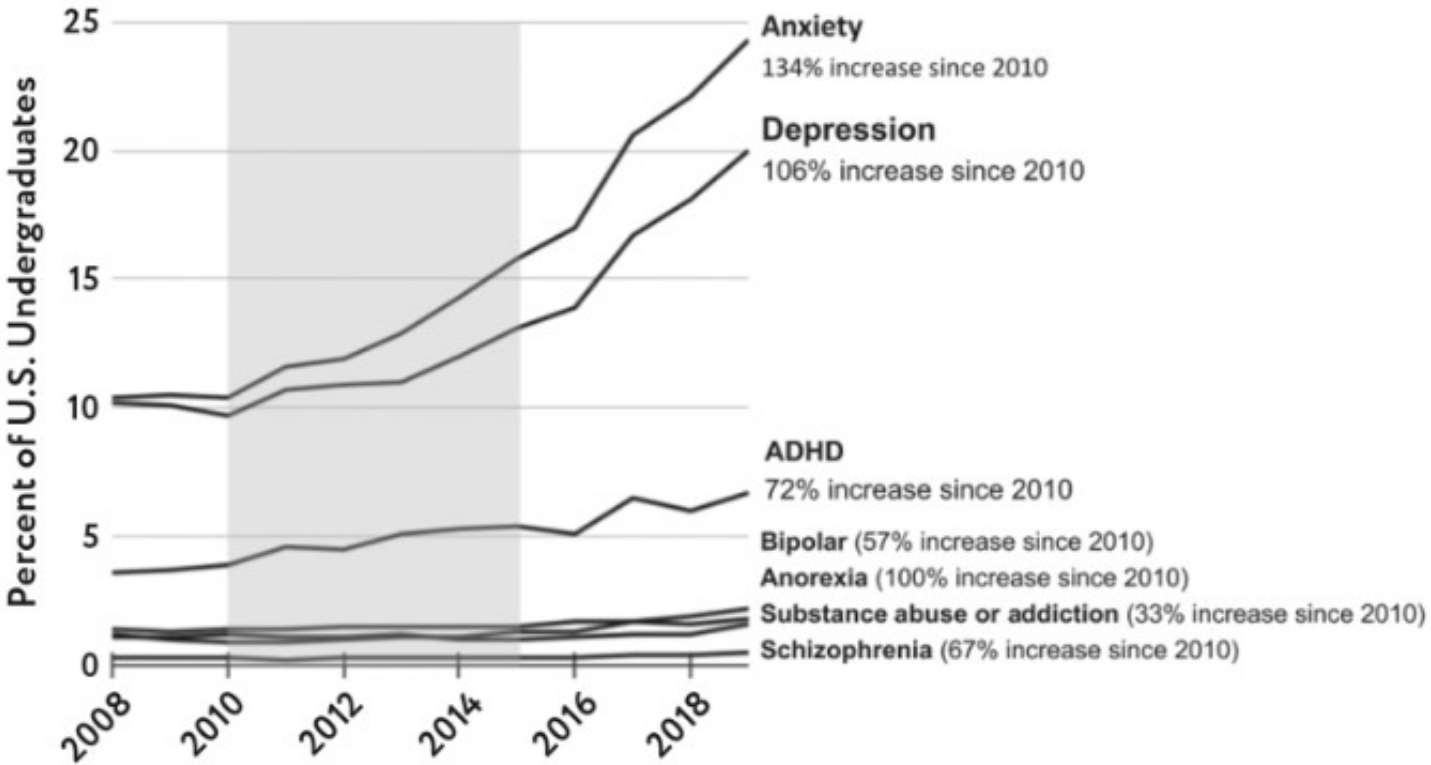
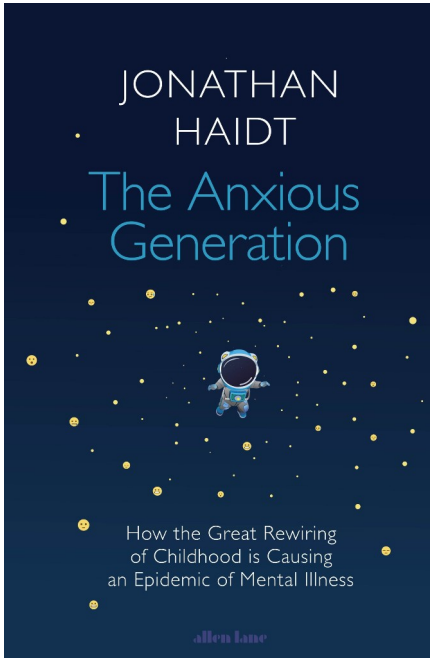


Figure 1.2. Percent of U.S. undergraduates with each of several mental illnesses. Rates of diagnosis of various mental illnesses increased in the 2010s among college students, especially for anxiety and depression. (Source: American College Health Association.)[9]



"The Anxious Generation", by Jonathan Haidt

Narcotics for the masses

“Cognitive Debt”



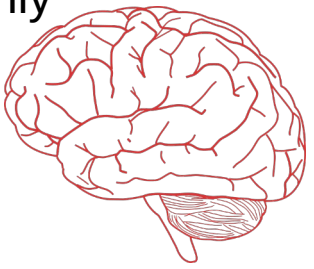
LLM





Search engine

Brain only



“Brain connectivity systematically scaled down with the amount of external support: the Brain-only group exhibited the strongest, widest-ranging networks, Search Engine group showed intermediate engagement, and **LLM assistance elicited the weakest overall coupling**. In session 4, LLM-to-Brain participants showed **weaker neural connectivity** and under-engagement of alpha and beta networks”

Kosmyna, Nataliya, et al. "Your brain on ChatGPT: Accumulation of cognitive debt when using an AI assistant for essay writing task." arXiv preprint arXiv:2506.08872 4 (2025). [link]

How does it work: Gen-AI / Fake Godmen

1) Has read a lot
    a) but without critical understanding
2) You can ask any Q on anything
    a) A could be wrong, but presumed correct
3) Feedback: difficult
4) Objective: max num seeking advise
    a) NOT whether advise is effective
5) Initial service free, but you lose money soon

